

Insulation Wear: PVC vs. XLPE

This chart shows the exponential decay in relative insulation resistance as operating temperature increases. The decay coefficients are adjusted to show a more pronounced curve, especially beyond the maximum operating temperatures. Note how XLPE’s resistance decays slower and it remains functional at a higher maximum temperature.

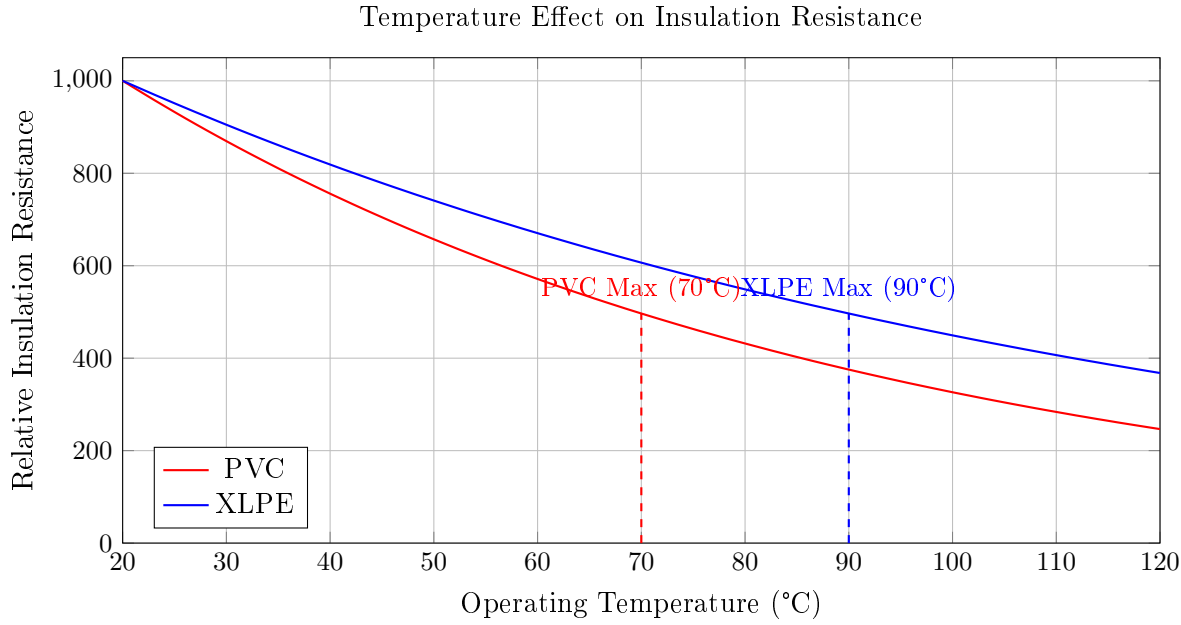


Figure 1: Comparison of insulation resistance decay for PVC and XLPE (More Curved Decay Model).

Economic Comparison of Conductors

These charts illustrate the trade-off between initial cost and estimated 30-year operating cost for different conductor materials.

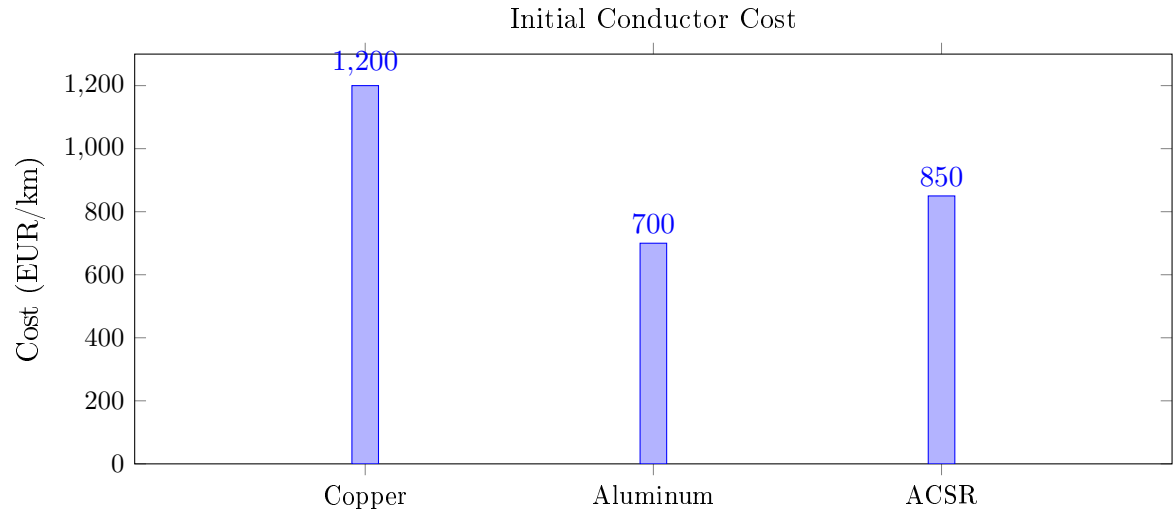


Figure 2: Initial material cost per kilometer.

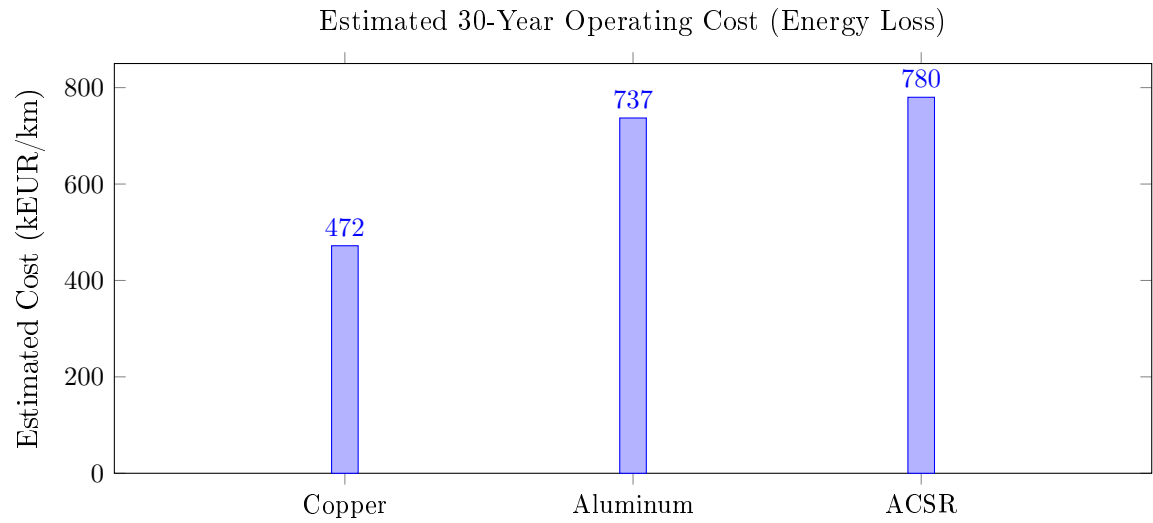


Figure 3: Estimated 30-year operating cost due to energy losses, based on assumed load (100A), area (50mm²), temp (50°C), and price (€0.15/kWh). For comparative purposes only.²